ACADEMIC CV

Pinglu **Zhang**

PhD Student in Computer Science and Technology
Institute of Fundamental and Frontier Sciences
University of Electronic Science and Technology of China
Supervisor **Prof. Quan Zou**

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RESEARCH PROFILE

My research focuses on algorithms for sequence and genome alignment, with an emphasis on scalable multiple sequence alignment and the comparison of complex repetitive regions, including centromeres. My current research addresses multiple-genome alignment and the construction of large-scale pangenome graphs.

Research interests Multiple sequence alignment; multiple-genome alignment; whole-genome alignment; centromeric and repetitive-region alignment; large-scale pangenome graph construction.

EDUCATION AND RESEARCH TRAINING

2025–Present **PhD Studies in Computer Science and Technology**
Institute of Fundamental and Frontier Sciences, UESTC.
Integrated Master's–PhD track, entered in 2023. Supervisor: Prof. Quan Zou.

2025–Present **Joint Doctoral Training**
Zhongguancun Academy, Beijing, China.

2024–2025 **Joint Graduate Training**
Yangtze Delta Region Institute (Quzhou), UESTC, Quzhou, China.

SELECTED RESEARCH CONTRIBUTIONS

HAlign-G: rapid and low-memory multiple-genome aligner for large-scale closely related genomes

Co-first author · Genome Biology 2025 / Code

Large-scale multiple-genome alignment using BWT-FM-LIS indexing, optimized K-band alignment and a star-alignment strategy.

Fast sequence alignment for centromeres with RaMA

First author · Genome Research 2025 / Code

Alignment of centromeric regions using rare-match anchors and wavefront alignment with a two-piece affine gap cost.

FMAlign2: a novel fast multiple nucleotide sequence alignment method for ultralong datasets

First author · Bioinformatics 2024 / Code

Multiple alignment of ultralong nucleotide sequences using suffix-array seeding and a vertical partitioning strategy.

RESEARCH FUNDING AND HONORS

2025	**NSFC Basic Research Program for Young Students** Doctoral Students program, National Natural Science Foundation of China. Project: Large-scale whole-genome alignment across multiple species.
Oct 2026	National Scholarship
Oct 2026	Outstanding Poster Award, 5th National Joint Meeting of Deans of Institutes for Frontier and Interdisciplinary Research
2026	Academic Rising Star, UESTC
2025	Outstanding Graduate Student and Academic Young Talent, UESTC
2023	First Prize (Ranked 1st Overall), CBC Data Challenge

Publications

Pinglu Zhang is shown in bold. † Equal contribution as co-first authors.

Sequence Alignment

01 **HAlign-G: rapid and low-memory multiple-genome aligner for large-scale closely related genomes**
Pinglu Zhang†, Tong Zhou†, Yanming Wei, Qinzhong Tian, Yixiao Zhai, Yizheng Wang, Quan Zou, Furong Tang, and Ximei Luo
Genome Biology · 2025 · 26: 406
doi: 10.1186/s13059-025-03881-3

02 **Fast sequence alignment for centromeres with RaMA**
Pinglu Zhang, Yanming Wei, Qinzhong Tian, Quan Zou, and Yansu Wang
Genome Research · 2025 · 35(5): 1209–1218
doi: 10.1101/gr.279763.124

03 **FMAlign2: a novel fast multiple nucleotide sequence alignment method for ultralong datasets**
Pinglu Zhang, Huan Liu, Yanming Wei, Yixiao Zhai, Qinzhong Tian, and Quan Zou
Bioinformatics · 2024 · 40(1): btae014
doi: 10.1093/bioinformatics/btae014

04 **High-Quality Large-Scale Viral Genome Multiple Sequence Alignment with Updated HAlign4**
Pinglu Zhang, Qinzhong Tian, Yixiao Zhai, Haodong Liu, and Quan Zou
International Conference on Intelligent Computing (ICIC 2026) · 2027 · LNCS 16671, 291–302. Published online 14 July 2026.
doi: 10.1007/978-981-92-3498-1_25

05 **Chimera: Ultrafast and Memory-efficient Database Construction for High-Accuracy Taxonomic Classification in the Age of Expanding Genomic Data**
Qinzhong Tian†, **Pinglu Zhang**†, Yanming Wei, Quan Zou, Yansu Wang, and Ximei Luo
bioRxiv · 2025 · Preprint
doi: 10.1101/2025.03.26.645388

06 **Fast and memory efficient partial order alignment with minipoa**
Haodong Liu, **Pinglu Zhang**, Yanming Wei, Qinzhong Tian, Yixiao Zhai, Quan Zou, and Mengting Niu
Genome Research · 2026 · 36(9): 1849–1862
doi: 10.1101/gr.282046.126

07 **deMEM: a novel divide-and-conquer framework based on de Bruijn graph for scalable multiple sequence alignment**
Yanming Wei, Zhaoyang Huang, **Pinglu Zhang**, Yizheng Wang, Yan Li, Liang Yu, and Quan Zou
GigaScience · 2026 · 15: giaf163
doi: 10.1093/gigascience/giaf163

08 **Application and Comparison of Machine Learning and Database-Based Methods in Taxonomic Classification of High-Throughput Sequencing Data**
Qinzhong Tian, **Pinglu Zhang**, Yixiao Zhai, Yansu Wang, and Quan Zou
Genome Biology and Evolution · 2024 · 16(5): evae102
doi: 10.1093/gbe/evae102

09 **TPMA: A two pointers meta-alignment tool to ensemble different multiple nucleic acid sequence alignments**
Yixiao Zhai, Jiannan Chao, Yizheng Wang, **Pinglu Zhang**, Furong Tang, and Quan Zou
PLOS Computational Biology · 2024 · 20(4): e1011988
doi: 10.1371/journal.pcbi.1011988

10 **ReAlign-P: a vertical iterative realignment method for protein multiple sequence alignment**
Yixiao Zhai, **Pinglu Zhang**, Quan Zou, and Ximei Luo
Bioinformatics · 2025 · 41(8): btaf421
doi: 10.1093/bioinformatics/btaf421

11 **HAlign 4: a new strategy for rapidly aligning millions of sequences**
Tong Zhou, **Pinglu Zhang**, Quan Zou, and Wu Han
Bioinformatics · 2024 · 40(12): btae718
doi: 10.1093/bioinformatics/btae718

12 **TAXICF: Efficient Index Construction and Accurate Metagenomic Classification with Interleaved Cuckoo Filters**
Qinzhong Tian, **Pinglu Zhang**, and Quan Zou
International Conference on Intelligent Computing (ICIC 2026) · 2027 · LNCS 16671, 349–360. Published online 14 July 2026.
doi: 10.1007/978-981-92-3498-1_30

13 **ReAlign-Star: An Optimized Realignment Method for Multiple Sequence Alignment, Targeting Star Algorithm Tools**
Yixiao Zhai, **Pinglu Zhang**, Yi Liu, and Quan Zou
International Conference on Intelligent Computing (ICIC 2025) · 2025 · LNCS 15867, 447–457.
doi: 10.1007/978-981-95-0030-7_38

Other Publications

14 **PS-Mixer: A Polar-Vector and Strength-Vector Mixer Model for Multimodal Sentiment Analysis**
Han Lin†, **Pinglu Zhang**†, Jiading Ling, Zhenguo Yang, Lap Kei Lee, and Wenyin Liu
Information Processing & Management · 2023 · 60(2): 103229
doi: 10.1016/j.ipm.2022.103229

15 **MMSpa is a deep learning-based tool that enhances the identification of spatial domains in spatial transcriptomics studies**
Yi Liu, Yixiao Zhai, **Pinglu Zhang**, Quan Zou, and Ximei Luo
PLOS Biology · 2026 · 24(1): e3003580
doi: 10.1371/journal.pbio.3003580

16 **A unified framework for potency-oriented AMP discovery via multi-modal learning and guided sequence synthesis**
Wenyu Zhang, Yizheng Wang, Yixiao Zhai, **Pinglu Zhang**, Yijie Ding, and Quan Zou
PLOS Computational Biology · 2026 · 22(9): e1014771
doi: 10.1371/journal.pcbi.1014771

17 **TCM@MPXV: A Resource for Treating Monkeypox Patients in Traditional Chinese Medicine**
Xin Zhang†, Feiran Zhou†, **Pinglu Zhang**, Quan Zou, and Ying Zhang
Current Bioinformatics · 2025 · 20(6): 557–563
doi: 10.2174/0115748936299878240723044438